



QR Code for
report verification

Case ID : 2600133939
Name : **MRS. SAROJ DEVI (OQG2604250052)**
Sex/Age : Female/64 Years
Bill. Loc. : OncQuest Laboratories Ltd., New Delhi
Ref. By :
Indication :

Sample Type : BLOOD IN CCF DNA TUBE
Date & Time Collected : 25-Apr-2026 07:57 AM
Date & Time Received : 27-Apr-2026 12:14 PM
Date & Time Reported : 11-May-2026 06:24 PM
Report Version : 1

Breast and Ovarian Extended Panel-Liquid Biopsy Assay

Clinical Indication:

Carcinoma Breast

Positive

(Clinically significant variant detected)

Key Findings:

Gene & Transcript	Variant	Exon	Coverage / VAF	Classification	Relevant Biomarker
PIK3CA NM_006218.4	c.1258T>C (p.Cys420Arg)	Exon 8	465x / 5.81%	Tier IA	Targeted therapies, guidelines & global clinical trials info attached

*Genetic test results are reported based on the recommendations of AMP-ASCO-CAP guidelines.

Test Description:

The Breast and Ovarian Cancer Extended - Liquid Biopsy Assay is an NGS panel based on molecular tagging technical method that tested to detect low MAFs mutation from plasma samples.

Variant Classification as per AMP-ASCO-CAP Guidelines:

Variant	A change in a gene. This could be disease causing (pathogenic) or not disease causing (benign).
Tier I: Variants of Strong Clinical Significance	Level A , biomarkers that predict response or resistance to US FDA-approved therapies for a specific type of tumor or have been included in professional guidelines as therapeutic, diagnostic, and/or prognostic biomarkers for specific types of tumors; Level B , biomarkers that predict response or resistance to a therapy based on well-powered studies with consensus from experts in the field, or have diagnostic and/or prognostic significance of certain diseases based on well-powered studies with expert consensus.
Tier II: Variants of Potential Clinical Significance	Level C , biomarkers that predict response or resistance to therapies approved by FDA or professional societies for a different tumor type (ie, off-label use of a drug), serve as inclusion criteria for clinical trials, or have diagnostic and/or prognostic significance based on the results of multiple small studies; Level D , biomarkers that show plausible therapeutic significance based on preclinical studies or may assist disease diagnosis and/or prognosis themselves or along with other biomarkers based on small studies or multiple case reports with no consensus.

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MC-7414



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Tier III: Variants of unknown clinical significance	Not observed at a significant allele frequency in the general or specific subpopulation or pancancer or tumor specific variant databases. No convincing published evidence of cancer Association
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Test Methodology:

- The cell-free DNA (cell-free total nucleic acid; cfTNA) isolated from the plasma fraction of whole blood.
- A single pool of multiplex PCR primers used to preparation of libraries from cell-free DNA (cfDNA) obtained from the plasma fraction of a single tube of whole blood. In process quality controls were determined for prepared library.
- The libraries were sequenced at minimum mean depth: >20,000x on Illumina Novaseq 6000 next generation sequencing platform.
- Highly automated workflow and streamlined bioinformatics analysis pipeline were used to provide a SNVs, Indels, CNVs and as part of the streamlined bioinformatics workflow, the mutational signature plot were generated.

Table: The Breast and Ovarian Cancer Extended Panel Gene List.

Gene List (93 Genes)

ACVR1B, CDKN2A, GEN1, NCOR1, SEPT9, AKT1, CHEK2, HERC1, NEK2, SMAD4, APC, CSMD1, HOXB13, NF1, SMARCA4, AR, CTNNB1, IRAK4, PALB2, STK11, ATM, DIRAS3, ITCH, PALLD, SYNE1, ATR, EGFR, KMT2C, PBRM1, TGFB1, AXIN2, EP300, KRAS, PCGF2, TP53, BAP1, EPCAM, MAP2K4, PIK3CA, TRAF5, BARD1, ERBB2, MAP3K1, PIK3R1, VHL, BLM, ERBB3, MDM2, PMS1, WEE1, BMPR1A, ERCC4, MED12, PMS2, XRCC2, BRCA1, ESR1, MEN1, PPM1L, XRCC3, BRCA2, EXOC2, MLH1, PTEN, ZBED4, BRIP1, EXT2, MRE11A, PTGFR, CASP8, FAM175A, MSH2, RAD50, CBFB, FANCC, MSH6, RAD51, CCND1, FBXO32, MUC16, RAD51C, CDH1, FGFR1, MUTYH, RAD51D, CDK4, FGFR2, MYC, RB1, CDK6, GATA3, NBN, RET

Disclaimer:

- This test is limited to listed genes analysis only, It should be noted that this test does not sequence all bases in a human genome, not all variants have been identified or interpreted, and this report is limited only to variants with evidence for causing or contributing to disease/clinical details provided to SN Gene Lab Pvt Ltd.
- This assay is not meant to interrogate most promoter regions, deep intronic regions, or other regulatory elements, and does not detect single or multi-exon deletions or duplications.
- Due to poor quality of cell-free total nucleic acid; cfTNA possibility of assay failure/compromised results that include low gene coverage and low variant depth, cannot be ruled out
- No Guarantee: By providing drug and clinical trial information for the reported diagnosis, SN GeneLab Pvt. Ltd. is not guaranteeing that any drug or clinical trial is necessarily appropriate for this patient. Healthcare providers should evaluate and interpret the information provided in this report, along with all other available clinical information about this patient, to determine the best treatment decisions in their own independent medical judgment. Patient management decisions should not be based on a single test, including this one, nor solely on the information contained in this report.
- Drugs: The drugs listed on the report are not ranked in any specific order as to predicted efficacy or appropriateness

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for this patient. SN GeneLab Pvt. Ltd. makes no guarantee or promise as to the effectiveness or suitability (or lack thereof) of any drug listed on this report. For more detailed information, healthcare providers should refer to the package insert for each FDA-approved drug listed in this report, and go to clinicaltrials.gov for information regarding drugs in clinical trials.

References:

1. Li, MM, et al. Standards and Guidelines for the Interpretation and Reporting of Sequence Variants in Cancer: A Joint Consensus Recommendation of the Association for Molecular Pathology, American Society of Clinical Oncology, and College of American Pathologists. *Journal of Molecular Diagnostics*. 19.1 (2017): 4-23
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12. R. Chaudhary et al. A scalable solution for tumor mutational burden from formalin-fixed, paraffin-embedded samples using the OncoPrint Tumor Mutation Load Assay. *Translational lung cancer research*, 2018.

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13. Li, H, and R, Durbin. Fast and accurate long-read alignment with Burrows-Wheeler transform. *Bioinformatics*. 26.5: (2010): 589-95.

----- End Of Report -----

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MC-7414



Sample Cancer Type: Breast Cancer

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Relevant Biomarkers

Tier	Genomic Alteration	Relevant Therapies (In this cancer type)	Relevant Therapies (In other cancer type)	Clinical Trials
IA	PIK3CA p.(C420R) c.1258T>C	inavolisib + palbociclib + hormone therapy ¹ alpelisib + hormone therapy ^{1, 2} capivasertib + hormone therapy ^{1, 2}	None*	29

* Public data sources included in relevant therapies: FDA¹, NCCN, EMA², ESMO
Tier Reference: Li et al. Standards and Guidelines for the Interpretation and Reporting of Sequence Variants in Cancer: A Joint Consensus Recommendation of the Association for Molecular Pathology, American Society of Clinical Oncology, and College of American Pathologists. J Mol Diagn. 2017 Jan;19(1):4-23.

Relevant Therapy Summary

In this cancer type

In other cancer type

In this cancer type and other cancer types

No evidence

PIK3CA p.(C420R) c.1258T>C

Relevant Therapy	FDA	NCCN	EMA	ESMO	Clinical Trials*
alpelisib + fulvestrant					
capivasertib + fulvestrant					
inavolisib + palbociclib + fulvestrant					
alpelisib, hormone therapy					(IV)
gedatolisib, palbociclib, hormone therapy, alpelisib					(III)
inavolisib, hormone therapy, palbociclib					(III)
alpelisib + hormone therapy					(II)
alpelisib, hormone therapy, dapagliflozin					(II)
everolimus					(II)
everolimus + chemotherapy					(II)
everolimus, hormone therapy					(II)
inavolisib					(II)
MEN-1611, chemotherapy					(II)
PIK3CA inhibitor, hormone therapy					(II)
alpelisib + elacestrant					(I/II)
evexomostat, alpelisib, hormone therapy					(I/II)
HTL-0039732, atezolizumab					(I/II)
JS-105, hormone therapy, dalpiciclib					(I/II)
afuresertib, hormone therapy					(I)
HH-CYH33, hormone therapy, palbociclib					(I)
HS-10352, hormone therapy					(I)
inavolisib, atezolizumab					(I)
inavolisib, chemotherapy, trastuzumab, pertuzumab					(I)
IPN-60090, capivasertib					(I)
JS-105					(I)
palbociclib, gedatolisib					(I)
RLY-2608, hormone therapy, palbociclib, ribociclib, PF-07220060					(I)
SNV-4818, hormone therapy					(I)

* Most advanced phase (IV, III, II/III, II, I/II, I) is shown and multiple clinical trials may be available.

For queries please call on 9586177755

Relevant Therapy Summary (continued)

☒ In this cancer type

☐ In other cancer type

☒ In this cancer type and other cancer types

☒ No evidence

PIK3CA p.(C420R) c.1258T>C (continued)

Relevant Therapy	FDA	NCCN	EMA	ESMO	Clinical Trials*
TOS-358	×	×	×	×	☒ (I)

* Most advanced phase (IV, III, II/III, II, I/II, I) is shown and multiple clinical trials may be available.

Relevant Therapy Details

Current FDA Information

☒ In this cancer type

☐ In other cancer type

☒ In this cancer type and other cancer types

FDA information is current as of 2025-05-14. For the most up-to-date information, search www.fda.gov.

PIK3CA p.(C420R) c.1258T>C

☒ alpelisib + fulvestrant

Cancer type: Breast Cancer

Label as of: 2024-01-18

Variant class: PIK3CA C420R mutation

Other criteria: ERBB2 negative, Hormone receptor positive

Indications and usage:
PIQRAY® is a kinase inhibitor indicated in combination with fulvestrant for the treatment of adults with hormone receptor (HR)-positive, human epidermal growth factor receptor 2 (HER2)-negative, PIK3CA-mutated, advanced or metastatic breast cancer as detected by an FDA-approved test following progression on or after an endocrine-based regimen.

Reference:
https://www.accessdata.fda.gov/drugsatfda_docs/label/2024/212526s009lbl.pdf

☒ capivasertib + fulvestrant

Cancer type: Breast Cancer

Label as of: 2025-02-13

Variant class: PIK3CA C420R mutation

Other criteria: ERBB2 negative, Hormone receptor positive

Indications and usage:
TRUQAP® is a kinase inhibitor indicated, in combination with fulvestrant for the treatment of adult patients with hormone receptor (HR)-positive, human epidermal growth factor receptor 2 (HER2)-negative, locally advanced or metastatic breast cancer with one or more PIK3CA/AKT1/PTEN-alterations as detected by an FDA-approved test following progression on at least one endocrine-based regimen in the metastatic setting or recurrence on or within 12 months of completing adjuvant therapy.

Reference:
https://www.accessdata.fda.gov/drugsatfda_docs/label/2025/218197s002lbl.pdf

PIK3CA p.(C420R) c.1258T>C (continued)

● inavolisib + palbociclib + fulvestrant

Cancer type: Breast Cancer

Label as of: 2024-10-10

Variant class: PIK3CA C420R mutation

Other criteria: ERBB2 negative, Hormone receptor positive

Indications and usage:
ITOVEBI™ is a kinase inhibitor indicated in combination with palbociclib and fulvestrant for the treatment of adults with endocrine-resistant, PIK3CA-mutated, hormone receptor (HR)-positive, human epidermal growth factor receptor 2 (HER2)-negative, locally advanced or metastatic breast cancer, as detected by an FDA-approved test, following recurrence on or after completing adjuvant endocrine therapy.

Reference:
https://www.accessdata.fda.gov/drugsatfda_docs/label/2024/219249s000lbl.pdf

● inavolisib + palbociclib + fulvestrant

Cancer type: Breast Cancer

Label as of: 2025-04-23

Variant class: PIK3CA C420R mutation

Other criteria: ERBB2 negative, Hormone receptor positive

Indications and usage:
IBRANCE® is a kinase inhibitor indicated:

- for the treatment of adult patients with hormone receptor (HR)-positive, human epidermal growth factor receptor 2 (HER2)-negative advanced or metastatic breast cancer in combination with:
 - an aromatase inhibitor as initial endocrine-based therapy; or
 - fulvestrant in patients with disease progression following endocrine therapy.
- in combination with inavolisib and fulvestrant for the treatment of adult patients with endocrine-resistant, PIK3CA-mutated, HR-positive, HER2-negative, locally advanced or metastatic breast cancer, as detected by an FDA-approved test, following recurrence on or after completing adjuvant endocrine therapy.

Reference:
https://www.accessdata.fda.gov/drugsatfda_docs/label/2025/207103s020lbl.pdf

Current NCCN Information

☒ In this cancer type ☐ In other cancer type ☒ In this cancer type and other cancer types

NCCN information is current as of 2025-05-01. To view the most recent and complete version of the guideline, go online to NCCN.org.
For NCCN International Adaptations & Translations, search www.nccn.org/global/what-we-do/international-adaptations.

Some variant specific evidence in this report may be associated with a broader set of alterations from the NCCN Guidelines. Specific variants listed in this report were sourced from approved therapies or scientific literature. These therapeutic options are appropriate for certain population segments with cancer. Refer to the NCCN Guidelines® for full recommendation.

All guidelines cited below are referenced with permission from the NCCN Clinical Practice Guidelines in Oncology (NCCN Guidelines®) National Comprehensive Cancer Network, Inc. 2023. All rights reserved. NCCN makes no warranties regarding their content.

PIK3CA p.(C420R) c.1258T>C

● alpelisib + fulvestrant

Cancer type: Breast CancerVariant class: PIK3CA C420R mutation

Other criteria: ERBB2 negative, Hormone receptor positive

NCCN Recommendation category: 1

Population segment (Line of therapy):

- Stage IV; Recurrent, Unresectable, Local, Regional, Invasive, Progression (Second-line therapy, Subsequent therapy); Preferred intervention

Reference: NCCN Guidelines® - NCCN-Breast Cancer [Version 4.2025]

● capivasertib + fulvestrant

Cancer type: Breast CancerVariant class: PIK3CA C420R mutation

Other criteria: ERBB2 negative, Hormone receptor positive

NCCN Recommendation category: 1

Population segment (Line of therapy):

- Stage IV; Recurrent, Unresectable, Local, Regional, Invasive, Progression (Second-line therapy, Subsequent therapy); Preferred intervention

Reference: NCCN Guidelines® - NCCN-Breast Cancer [Version 4.2025]

● inavolisib + palbociclib + fulvestrant

Cancer type: Breast CancerVariant class: PIK3CA C420R mutation

Other criteria: ERBB2 negative, Hormone receptor positive

NCCN Recommendation category: 1

Population segment (Line of therapy):

- Stage IV; Recurrent, Unresectable, Local, Regional, Invasive, Progression (First-line therapy); Useful in certain circumstances

Reference: NCCN Guidelines® - NCCN-Breast Cancer [Version 4.2025]

For queries please call on 9586177755

Current EMA Information

- ☒ In this cancer type
- ☐ In other cancer type
- ☒ In this cancer type and other cancer types

EMA information is current as of 2025-05-14. For the most up-to-date information, search www.ema.europa.eu.

PIK3CA p.(C420R) c.1258T>C

☒ alpelisib + fulvestrant

Cancer type: Breast CancerLabel as of: 2025-04-30Variant class: PIK3CA C420R mutation

Other criteria: ERBB2 negative, Hormone receptor positive

Reference:
https://www.ema.europa.eu/en/documents/product-information/piqray-epar-product-information_en.pdf

☒ capivasertib + fulvestrant

Cancer type: Breast CancerLabel as of: 2025-03-07Variant class: PIK3CA C420R mutation

Other criteria: ERBB2 negative, Hormone receptor positive

Reference:
https://www.ema.europa.eu/en/documents/product-information/truqap-epar-product-information_en.pdf

Current ESMO Information

- ☒ In this cancer type
- ☐ In other cancer type
- ☒ In this cancer type and other cancer types

ESMO information is current as of 2025-05-01. For the most up-to-date information, search www.esmo.org.

PIK3CA p.(C420R) c.1258T>C

● alpelisib + fulvestrant

Cancer type: Breast Cancer

Variant class: PIK3CA C420R mutation

Other criteria: ERBB2 negative, ER positive

ESMO Level of Evidence/Grade of Recommendation: I / B

Population segment (Line of therapy):

- Luminal A; Advanced, Metastatic (Second-line therapy); ESMO-MCBS v1.1 score: 2

Reference: ESMO Clinical Practice Guidelines - ESMO-Metastatic Breast Cancer [Ann Oncol (2021) VOLUME 32, ISSUE 12, P1475-1495, DECEMBER 01, 2021; DOI:<https://doi.org/10.1016/j.annonc.2021.09.019>]



Clinical Trials Summary

PIK3CA p.(C420R) c.1258T>C

NCT ID	Title	Phase
NCT05392608	SEQUEL-Breast: A Phase II study on the combination of alpelisib and fulvestrant after progression on previous therapy with fulvestrant in patients with PIK3CA-mutated, hormone-receptor positive, HER2 negative advanced breast cancer	II
NCT05025735	Alpelisib, Fulvestrant and Dapagliflozin for the Treatment of HR+, HER2 -, PIK3CA Mutant Metastatic Breast Cancer	II
NCT06846099	A Phase I/Ib Study to Evaluate the Safety, Tolerability, Pharmacokinetics, Efficacy and Indications of PI3Kα-Selective Inhibitor RP903 (JS105) in Subjects with Advanced Malignancies	I
NCT05631795	ALPelisib INDia Safety STUDY (ALPINIST): A Phase IV, Prospective, Multicenter, Open-label, Non-comparative, Interventional Study to Assess the Safety of Alpelisib Plus Fulvestrant, in Men and Post-menopausal Women With HR Positive, HER2-negative, Advanced Breast Cancer (aBC) With a PIK3CA Mutation, Whose Disease Has Progressed on or After Endocrine Based Treatment	IV
NCT05486143	Phase III, Open-Label, Randomized, Study Comparing Gedatolisib Combined With Fulvestrant & With or Without Palbociclib to Standard-of-Care Therapies in Patients With HR-Positive, HER2-Negative Advanced Breast Cancer Previously Treated With a CDK4/6 Inhibitor in Combination w/Non-Steroidal Aromatase Inhibitor Therapy	III
NCT06790693	A Phase III, Multicenter, Randomized, Double-Blind, Placebo-Controlled Study Evaluating the Efficacy and Safety of Inavolisib Plus a CDK4/6 Inhibitor and Letrozole Versus Placebo Plus a CDK4/6 Inhibitor and Letrozole in Patients With Endocrine-Sensitive PIK3CA-Mutated, Hormone Receptor-Positive, HER2-Negative Advanced Breast Cancer	III
NCT05933395	Genetically-informed Therapy for ER+ Breast Cancer in a Post-CDK4/6 Inhibitor Setting: a Phase II Umbrella Study (GERTRUDE)	II
NCT04762979	A Phase II, Single Arm, Non-randomized Study of Alpelisib (BYL719) in Combination With Continued Endocrine Therapy Following Progression on Endocrine Therapy in Hormone Receptor Positive, HER2 Negative, PIK3CA Mutant Metastatic Breast Cancer	II
NCT05983107	A Prospective Cohort, Open, Phase II Clinical Study of Chidamide/Everolimus Combined With Endocrine Therapy for PIK3CA Wild-type/Mutant Hormone Receptor Positive/Human Epidermal Growth Factor Receptor 2 Negative Advanced Breast Cancer	II
NCT05332561	Genomics Guided Targeted Post-neoadjuvant Therapy in Patients With Early Breast Cancer - a Multicenter, Open-label, Umbrella Phase-II Study - COGNITION-GUIDE	II
NCT05810870	Phase II Study for Patients With Advanced Triple Negative or Metaplastic HR-positive/HER2-negative, PIK3CA/PTEN-altered Breast Cancer Treated With Eribulin in Combination With MEN1611	II
NCT05594095	Precision Platform Study of HR+/ HER2-advanced Breast Cancer Based on SNF Typing (A Prospective, Open-label, Multi-center, Phase II Platform Study)	II
NCT05563220	A Phase Ib/II, Open-Label Umbrella Study To Evaluate Safety And Efficacy Of Elacestrant In Various Combination In Patients With Metastatic Breast Cancer	I/II
NCT05455619	Study to Assess the Safety and Efficacy of Evexomostat (SDX-7320) in Combination With A PI3K or Akt Inhibitor Plus Fulvestrant in Patients With HR+, HER2-, Metastatic Breast Cancer With PI3K Pathway Alteration(s)	I/II
NCT06208410	A Clinical Study of JS105 in Combination With Other Anti-tumor Therapies in Patients With Advanced Solid Tumors	I/II
NCT04856371	A Multicenter, Open-label, Phase Ib Study to Evaluate the Safety, Tolerability, Pharmacokinetics and Preliminary Efficacy of CYH33 in Combination With Endocrine Therapy With or Without Palbociclib in Patients With PIK3CA Mutant, HR+, HER2- Advanced Breast Cancer.	I

Clinical Trials Summary (continued)

PIK3CA p.(C420R) c.1258T>C (continued)

NCT ID	Title	Phase
NCT05504213	A Phase Ib, Open-label, Multicenter Study to Evaluate Safety, Tolerability, Pharmacokinetics, and Efficacy of HS-10352 in Combination With Fulvestrant in Patients With Hormone Receptor (HR) Positive, Human Epidermal Growth Factor 2 (HER2)-Negative, PIK3CA Mutation, Locally Advanced or Metastatic Breast Cancer	I
No NCT ID	A Phase Ib, Open-label, Dose-escalation and Dose-expansion Study Evaluating the Safety, Tolerability, Pharmacokinetics, and Preliminary Antitumor Activity of Inavolisib in Combination with Paclitaxel in Patients with Locally Advanced or Metastatic Solid Tumors	I
NCT05216432	A First-in-Human Study of Mutant-selective PI3Kα Inhibitor, RLY-2608, As a Single Agent in Advanced Solid Tumor Patients and in Combination with Fulvestrant in Patients with Advanced Breast Cancer	I
NCT05683418	A Study to Evaluate the Safety and Tolerability of the Covalent Phosphoinositide-3-Kinase (PI3K)-α Inhibitor, TOS-358, in Adult Subjects with Select Solid Tumors	I
NCT04851613	A Phase Ib/III Study to Evaluate the Efficacy and Safety of Afuresertib Plus Fulvestrant in Patients With Locally Advanced or Metastatic HR+/HER2- Breast Cancer Who Failed Standard of Care Therapies	I
NCT04355858	Precision Treatment of Luminal Advanced Breast Cancer Based on Molecular Subtyping	II
NCT06736704	A Phase 1, Open-Label Dose Escalation and Expansion Study of SNV4818 As Monotherapy or in Combination with Other Anticancer Agents in Participants with Advanced Solid Tumors	I
NCT03239015	Efficacy and Safety of Precision Therapy in Refractory Tumor (Long March Pathway)	II
NCT06496568	A Phase I/Ib Study Evaluating Single-Agent Inavolisib and Inavolisib Plus Atezolizumab in PIK3CA-Mutated Cancers	I
NCT05039801	A Phase I Open-Label, Dose-Escalation and Dose-Expansion Study to Investigate the Safety, Pharmacokinetics, and Anti-Tumor Activity of IACS-6274 as Monotherapy and in Combination in Patients With Advanced Solid Tumors	I
No NCT ID	Phase I/Ib Clinical Study of the Safety, Tolerability, Pharmacokinetics and Preliminary Efficacy and Indication Cohort Expansion of the Selective PI3Kα Inhibitor RP903 (Also Known as JS105) in Patients with Advanced Malignancies	I
NCT03065062	Phase I Study of the CDK4/6 Inhibitor Palbociclib (PD-0332991) in Combination With the PI3K/mTOR Inhibitor Gedatolisib (PF-05212384) for Patients With Advanced Squamous Cell Lung, Pancreatic, Head & Neck and Other Solid Tumors	I
NCT05944237	A Cancer Research UK Phase I/IIa Trial of HTL0039732, Given Orally as Monotherapy and in Combination With Immunotherapy or Other Approved Therapies in Participants With Advanced Solid Tumours	I/II

Biomarker Descriptions

PIK3CA p.(C420R) c.1258T>C

phosphatidylinositol-4,5-bisphosphate 3-kinase catalytic subunit alpha

Background: The PIK3CA gene encodes the phosphatidylinositol-4,5-bisphosphate 3-kinase catalytic subunit alpha of the class I phosphatidylinositol 3-kinase (PI3K) enzyme¹. PI3K is a heterodimer that contains a p85 regulatory subunit, which couples one of four p110 catalytic subunits to activated tyrosine protein kinases^{2,3}. The p110 catalytic subunits include p110α, β, δ, γ and are encoded by genes PIK3CA, PIK3CB, PIK3CD, and PIK3CG, respectively². PI3K catalyzes the conversion of phosphatidylinositol (4,5)-bisphosphate (PI(4,5)P₂) into phosphatidylinositol (3,4,5)-trisphosphate (PI(3,4,5)P₃) while the phosphatase and tensin homolog (PTEN) catalyzes the reverse reaction^{4,5}. The reversible phosphorylation of inositol lipids regulates diverse aspects of cell growth and metabolism^{4,5,6,7}. Recurrent somatic alterations in PIK3CA are frequent in cancer and result in the activation of PI3K/AKT/MTOR pathway, which can influence several hallmarks of cancer including cell proliferation, apoptosis, cancer cell metabolism and invasion, and genetic instability^{8,9,10}.

Alterations and prevalence: Recurrent somatic activating mutations in PIK3CA are common in diverse cancers and are observed in 20-30% of breast, cervical, and uterine cancers and 10-20% of bladder, gastric, head and neck, and colorectal cancers^{11,12}. Activating mutations in PIK3CA commonly occur in exons 10 and 21 (previously referred to as exons 9 and 20 due to exon 1 being untranslated)^{13,14}. These mutations typically cluster in the exon 10 helical (codons E542/E545) and exon 21 kinase (codon H1047) domains, each having distinct mechanisms of activation^{15,16,17}. PIK3CA resides in the 3q26 cytoband, a region frequently amplified (10-30%) in diverse cancers including squamous carcinomas of the lung, cervix, head and neck, and esophagus, and in serous ovarian and uterine cancers^{11,12}.

Potential relevance: The PI3K inhibitor, alpelisib¹⁸, is FDA-approved (2019) in combination with fulvestrant for the treatment of patients with PIK3CA-mutated, hormone receptor (HR)-positive, human epidermal growth factor receptor 2 (HER2)-negative, advanced or metastatic breast cancer. Additionally, a phase Ib study of alpelisib with letrozole in patients with metastatic estrogen receptor (ER)-positive breast cancer showed the clinical benefit rate, defined as lack of disease progression ≥ 6 months, was 44% (7/16) in PIK3CA-mutated tumors and 20% (2/20) in PIK3CA wild-type tumors¹⁹. Specifically, exon 20 H1047R mutations were associated with more durable clinical responses in comparison to exon 9 E545K mutations¹⁹. However, alpelisib did not improve response when administered with letrozole in patients with ER+ early breast cancer with PIK3CA mutations²⁰. The FDA also approved the kinase inhibitor, capivasertib (2023)²¹ in combination with fulvestrant for locally advanced or metastatic HR-positive, HER2-negative breast cancer with one or more PIK3CA/AKT1/PTEN-alterations following progression after endocrine treatment. The kinase inhibitor, inavolisib²², is also FDA-approved (2024) in combination with palbociclib and fulvestrant for the treatment of adults with endocrine-resistant, PIK3CA-mutated, HR-positive, and HER2-negative breast cancer. Case studies with mTOR inhibitors sirolimus and temsirolimus report isolated cases of clinical response in PIK3CA mutated refractory cancers^{23,24}.

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